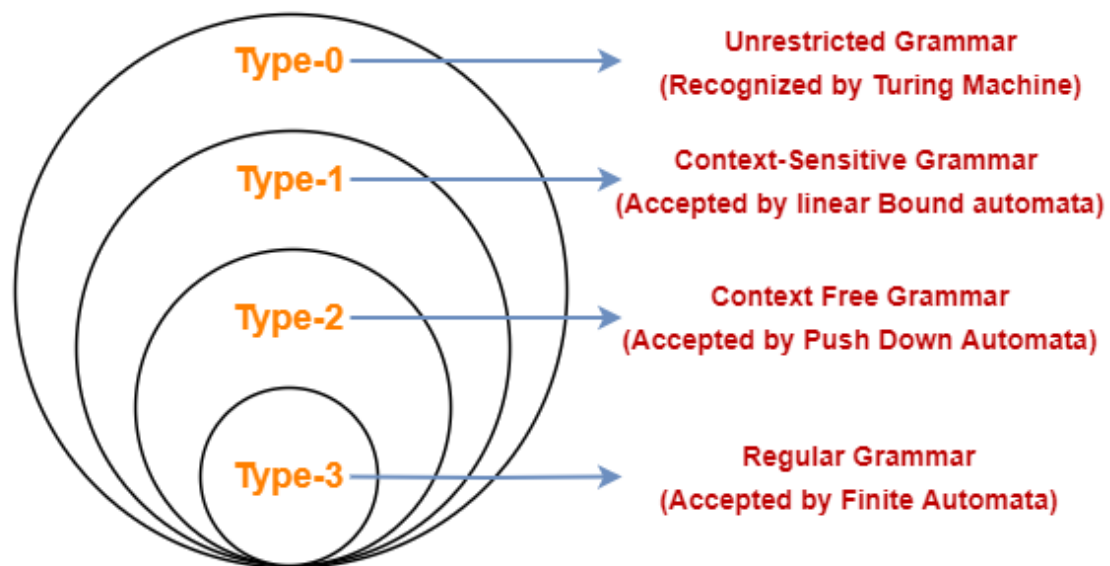


Grammars

Noam Chomsky described the grammar hierarchy in his work "Three models for the description of language". The hierarchy has four grammars:



Grammar	Language	Recognizing automaton	Rules	Examples
Type 3	Säännölliset	DFA, regular expression	$A \rightarrow a$ $A \rightarrow aB$ tai $A \rightarrow Ba$	$L = \{a^n \mid n > 0\}$
Type 2	Context-free	PDA (Push down automaton), Nondeterministic pushdown automaton	$A \rightarrow \alpha$	$L = \{a^n b^n \mid n > 0\}$
Type 1	Context-sensitive	LBA, Linearly bounded Nondeterministic Turing machine	$\alpha A \beta \rightarrow \alpha \gamma \beta$	$L = \{a^n b^n c^n \mid n > 0\}$
Type 0	Unrestricted grammar, Recursively enumerable	Turing machine (infinite tape)	$\gamma \rightarrow \alpha$	$L = \{w \mid w \text{ describes a finite turing machine}\}$

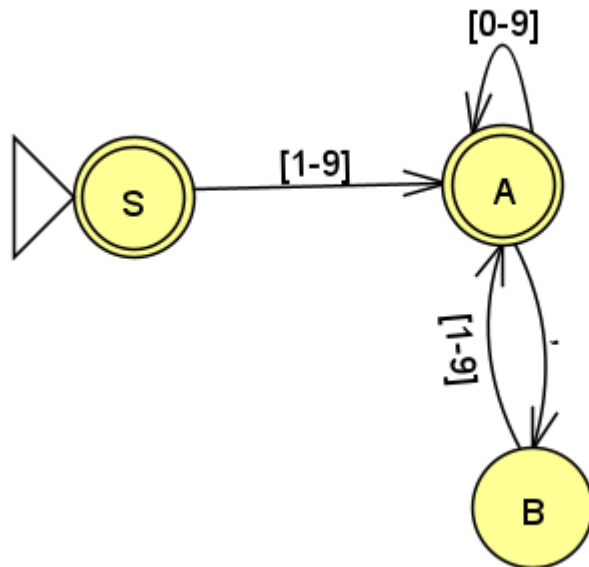
Capital letters represent nonterminals, lowercase letters terminals
Let the grammar be:

$S \rightarrow \lambda$ $S \rightarrow [1-9]A$

$A \rightarrow ,B$ $B \rightarrow [1-9]A$

$A \rightarrow [0-9]A$ $A \rightarrow \lambda$

Let's draw the corresponding automaton with JFLAP and convert it to a regular expression
(Convert $\rightarrow$ Convert FA to RE)



Which language does the automaton/regular expression accept?

Regular grammars

A type 3 grammar is defined as:

The empty language $\emptyset$ is a regular language

$\Sigma = \{a,b,c,d,\dots,z\}$ is a finite set of alphabets, which can be any characters

If A is a regular language, then A^* is also a regular language (Kleene star)

Thus, the empty string $\{\epsilon\}$ is also a regular language

If A and B are regular languages, $A \cup B$ (union) and $A \cdot B$ (concatenation) are regular languages

Other languages over the alphabet Σ are not regular languages

Kleene closure

Let V be a set of strings (1) or alphabets (a)

1) V^* is the set formed from V by concatenation and is closed under the concatenation operation. Thus, V is a subset of V^* and V^* is the smallest such set.

2) V^* is the set of all strings formed from the alphabets of set V .

Definition

$$V^0 = \{\varepsilon\}$$

$$V^1 = V$$

$$V^{i+1} = \{wv : w \in V^i, v \in V\}, i > 0$$

$$V^* = \bigcup_{i \geq 0} V^i = V^1 \cup \dots$$

For example

$\{ "ab", "c" \}^* = \{ \varepsilon, "ab", "c", "abab", "abc", "cab", "cc", "ababab", "ababc", "abcab", "abcc", "cabab", "cabcb", "ccab", "ccc", \dots \}$.

$$V^+ = \bigcup_{i \geq 1} V^i = V^1 \cup \dots$$

$\{ "a", "b", "c" \}^+ = \{ "a", "b", "c", "aa", "ab", "ac", "ba", "bb", "bc", "ca", "cb", "cc", "aaa", "aab", \dots \}$.

A regular language can be defined

- ➔ By a deterministic finite automaton
- ➔ By a nondeterministic finite automaton
- ➔ By a regular expression
- ➔ By a Turing machine
- ➔ ... and by a few other methods outside the scope of this course

Regular expression

A regular expression over an alphabet Σ is defined as

" Constants"

- ➔ $\emptyset$ i.e. the empty set
- ➔ ε i.e. the empty string
- ➔ $a \in \Sigma$, i.e. a letter in the alphabet Σ

Operations

- ➔ Concatenation: Let K and L be regular expressions over the alphabet (Σ sets of strings Σ^*), then KL is also a regular expression
- ➔ Choice: Let K and L be regular expressions over the alphabet (Σ sets of strings Σ^*), then $K|L$ is also a regular expression.

➔ Repetition or Kleene star K^* is the smallest superset that is closed under concatenation and also contains the empty string ϵ .

Precedence

- 1) Kleene $*$
- 2) Concatenation
- 3) Choice $|$

Some examples

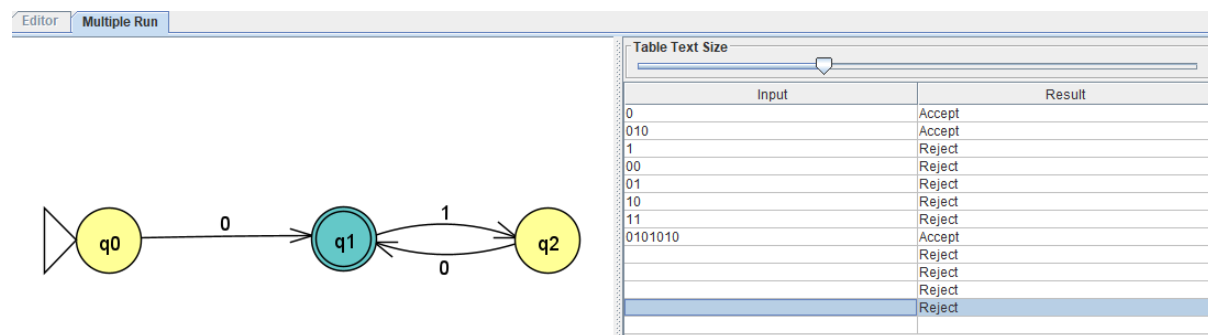
Grammar: $L = \{ 0(10)^n \mid n \geq 0 \}$

Regular expression: $0(10)^*$

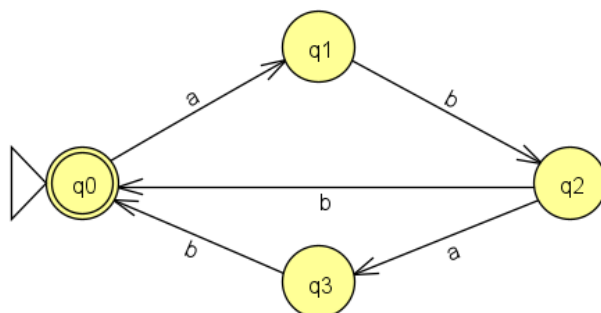
Grammar rules:

1. $S \rightarrow 0A$
2. $A \rightarrow 10A \mid \epsilon$

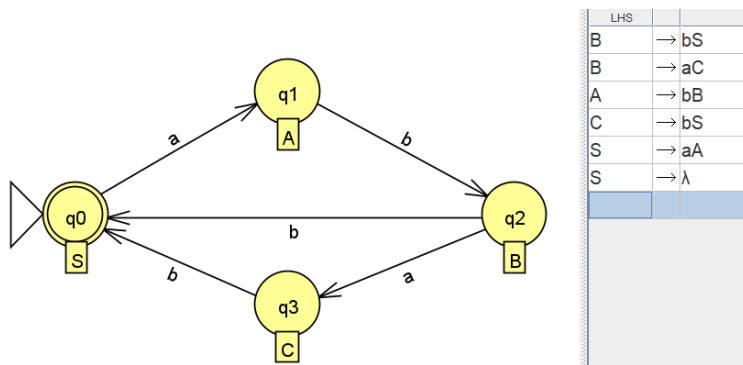
DFA:



What grammar is formed from this? And what about the regular expression?

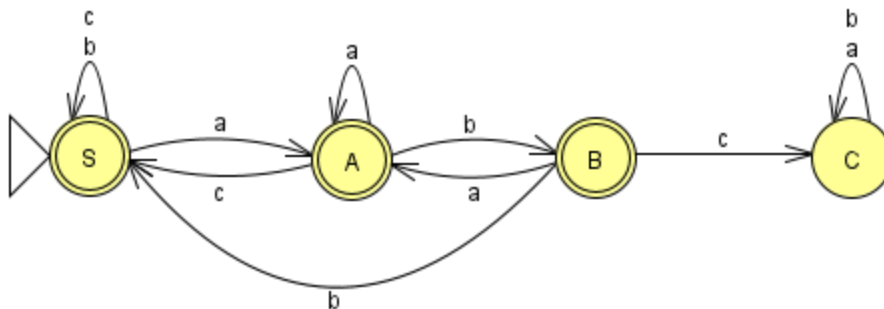


$(abb|abab)^*$



What kind of automaton would be generated from this grammar?

$S \rightarrow bS$ $A \rightarrow aA$ $B \rightarrow \lambda$ $C \rightarrow aC$
 $S \rightarrow cS$ $A \rightarrow cS$ $B \rightarrow cC$ $C \rightarrow bC$
 $S \rightarrow \lambda$ $A \rightarrow bB$ $B \rightarrow bS$
 $S \rightarrow aA$ $A \rightarrow \lambda$ $B \rightarrow aA$



Regular expression: $(b|c)^*a^*b$ (.... so it's not exactly very easy.

there are many languages easily described by a DFA that are not easily described by a regular expression. For instance, determining the validity of a given [ISBN](#) requires computing the modulus of the integer base 11, and can be easily implemented with an 11-state DFA. However, converting it to a regular expression results in a 2,14 megabytes file

Binary numbers divisible by two

$^(1^*0)^*\$$

Binary numbers divisible by four

$^((10)^*0|(10)^*11(1|0(10)^*11)^*0(10)^*0)^*\$$

Let's return to this.

Regular languages in practice

- 1) A character is a character, except
- 2) ., +, *, ?, ^, \$, (,), [,], {, }, |, \. which have a special meaning
- 3) \ which can make a special character into a character, e.g. \\$
- 4) \\ thus for the character \
- 5) \n \t \r newline, tab, carriage return
- 6) | means “or”
- 7) [aeiouyääö] is one consonant
- 8) [0-9] is some one number
- 9) [^] NOT, i.e. [^0-9] means any character except a number
- 10) ? one or none
- 11) ^ start \$ end

With these we get off to a good start, the list is by no means exhaustive, more will come up during the course.